KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

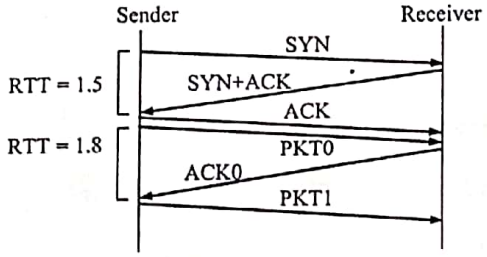
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is access network? Describe the enterprize access network with diagram. (08)
   b) How does data pass through network in case of circuit switching? Explain in brief. (08)
   c) What is troughput? What will be the troughput in case of Internet scenario? (06)
   d) What is the differences between RDT 2.1 and RDT 2.2? Draw RDT 2.2 finite state machine for sender and receiver. Briefly describe this. (13)

2. a) Why is UDP used in transport layer though it is unreliable? (05)
   b) In GB3, if every 5th packet is being transmitted is lost and if we have to send 10 packets, then how many transmissions are required? (11)
   c) What is the relation between window size and sequence number in case of selective repeat? Why is this relation needed? (10)
   d) Write short notes on the following terms: (09)
      (i) Demultiplexing, (ii) Characteristics of UDP, and (iii) UDP segment format.

3. a) What is "syn-flooding" attack in case of connection establishment in TCP? Explain. (06)
   b) How many ways are there to terminate a connection in TCP? Explain the ways in proper diagram. (12)
   c) Suppose receiver application program consumes data as 1byte at a time. For this, the receiver TCP announces the window size as 1byte. So, the sending TCP sends 1byte at a time and the traffic intensity will be high. What has to be done to solve the problem? Explain two solutions. (06)
   d) Explain Karn's algorithm in case of TCP timer. What will be the RTO of PKT1 in the following figure? (11)



4. a) How does TCP recognize the network congestion? Explain the Reno approach of managing this congestion by using necessary illustration. (13)
   b) Explain block cipher and cipher-block chain algorithm in symmetric key cryptography. Why will we use cipher-block chain instead of block cipher? (14)
   c) In case of RSA, if primes are $p = 5, q = 7$, public key is (35,5) and cipher text $c = 18$, then decrypt the cipher text to get plain text. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose two nodes start to transmit at the same time a packet of length L over a broadcast channel of rate R. Denote the propagation delay between the two nodes as d_{prop}. What there be a collision if $d_{prop} > \frac{L}{R}$? Why or why not. (08)
   b) Briefly describe that two dimensional parity checks can correct and detect a single bit error but a double bit error that can be detected but not corrected. (08)

c) Why is an ARP query sent within a broadcast frame? Why is an ARP repose sent within a frame with a specific MAC address? (07)

d) You have to look carefully Figure 5(d). Computer A wants to send a frame to computer C'. Describe the two switches' behavior when A connects to C' and after completing the communication, how the switch table of both switches look like? (Initially switch table was empty). Finally, write down the self-learning algorithm of switch. (12)

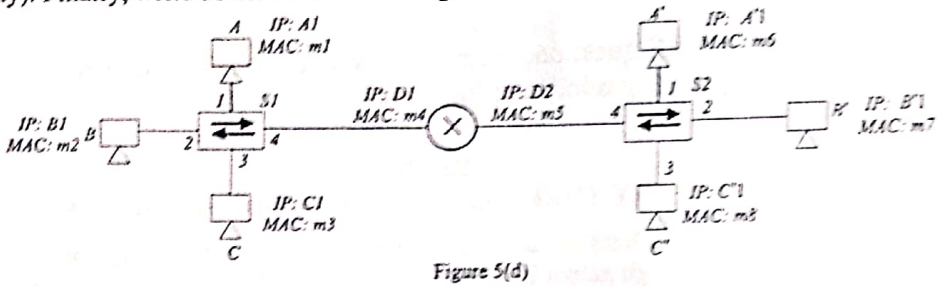


Figure 5(d)

6. a) Suppose you wanted to do a transaction securely from a remote client to a server as fast as possible. Would you use UDP or TCP? Why? (07)

b) What is meant by handshaking protocol? Why is it said that FTP sends control information "out of band"? (07)

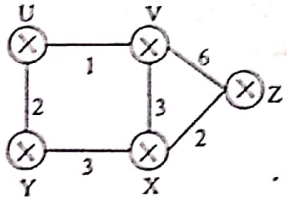
c) What are the advantages and disadvantages of a circular DHT (with no shortcuts)? (08)

d) Suppose the CSE department of KUET has 200 personal computer and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PC. How personal computer get the IP address if you have a network address 172.16.0.0/24. Also, state the dynamic range of IP addresses. (13)

7. a) Briefly describe different types of switching fabrics. Which, if any, can send multiple packets across the fabric in parallel? (06)

b) Define HOL blocking and longest prefix matching with appropriate example. (08)

c) Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12)



d) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (09)

8. a) What is NAT traversal problem? How can you solve it? Explain briefly with appropriate figure. (11)

b) Compare and contrast link-state and distance vector routing algorithms. (08)

c) Describe the technologies that are used for transitioning from IPv6 to IPv4 to IPv6. (07)

d) Consider the generator G = 10011 and suppose that D has the value 1010101010. What is the value of R? How can you detect the error using Cyclic Redundancy Check method? (09)